Movie Recommendation System

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Introduction

Problem

- Recommendation systems use a wide range of methods for extracting ratings and making recommendations.
- We wanted to create one of these systems based around predicting how a user would rate a movie they had not seen, not just presenting the highest average rating.
- We also wanted to compare simpler approaches against more complex ones

Hypothesis

The neural network will achieve the best performance because it can combine embeddings with both movie and user demographic features to produce a more accurate prediction.

Project Goal

Goal

- Predict a user's rating for a movie they have not seen.
- Compare different methods with different metrics used
- Evaluate the performance of the models using RMSE (Root Mean Square Error) and MAE (Mean Absolute Error)

Models Used

Average Baseline

- Predicts ratings using simple averages

KNN (K-Nearest Neighbor)

- Predicts ratings based on users that share similar rating patterns

SVD / Matrix Factorization

- Prediction based on learned patterns from a rating matrix

Linear Regression

- Makes predictions from the numeric features of users and movie average ratings

Neural Network

- Uses embeddings including extra user and movie features to learn complex patterns

How the Recommendation System Works

1. User

- A user has watched and rated some movies
- Ratings are on a scale of 1 to 5

2. Known Ratings + Features

- Past user ratings
- Movie genres
- User information (age, gender, occupation)

3. Models Predict Ratings

- Baseline
- KNN
- SVD (Matrix Factorization)
- Linear Regression
- Neural Network

4. Top Movie Recommendations

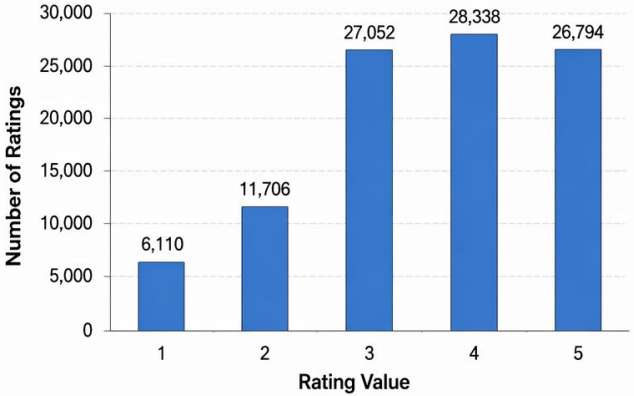
- Predict ratings for unseen movies
- Recommend movies with the highest predicted ratings

Dataset Overview: MovieLens 100k

Key Statistics

943	1,682	100,000	1 – 5
Users	Movies	Ratings	Rating Scale

Distribution of Ratings

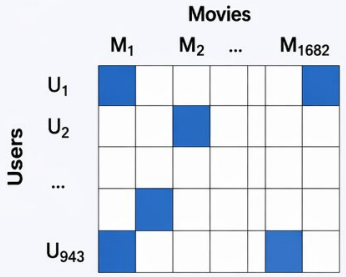


Dataset Files Used

u.data
User ratings for movies
(user_id, movie_id, rating, timestamp)

u.item
Movie information including genres
(movie_id, title, genres, etc.)

u.user
User demographics
(age, gender, occupation, zip code)



- Matrix is very sparse.
- Most users rate only a small fraction of movies.
- Predicting unseen ratings is the key challenge.

Dataset Files Used

01 u.data

- Contains movie ratings from users
- user_id movie_id rating timestamp
- 196 242 3 881250949

02 u.item

- Movie name and genres
- movie_id title Action Comedy...
- 1 Toy Story (1995) 0 1...

03 u.user

- User demographic data
- user_id age gender occupation zipcode
- 10 53 F lawyer 90703

Preprocessing

1. Load Files

- u.data
- u.item
- u.user

2. Merge Data

- ratings + movie info
- add user info for neural model

3. Train / Test Split

80% 20%

- 80% training data
- 20% testing data
- random_state = 35

4. Model-Specific Prep

- **Baseline:** user/movie averages
- **KNN & SVD:** user-movie matrix
- **Linear Regression:** summary features
- **Neural Net:** encoded IDs + genres + demographics

5. Evaluate Models

- RMSE
- MAE
- compare performance

$$\text{RMSE} = \sqrt{\frac{\sum (y_i - y_p)^2}{n}}$$

Evaluation Methods

$$\text{MAE} = \frac{|(y_i - y_p)|}{n}$$

y_i = actual value
y_p = predicted value
n = number of observations/rows

- We used RMSE and MAE to evaluate the models
- All models were tested with the same train/test split and random state.

RMSE

- Root Mean Square Error
- Measures how far predictions are from the actual ratings
- Larger errors = larger score penalization
- The lower the RMSE value, the better

MAE:

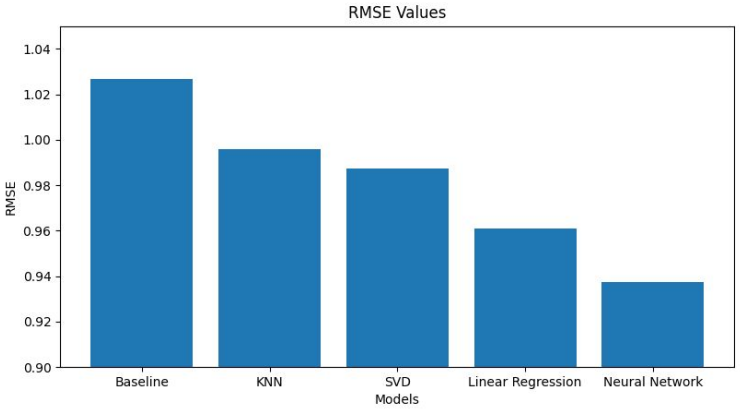
- Mean Absolute Error
- Measures the average size of a prediction error
- Lower MAE = better performance

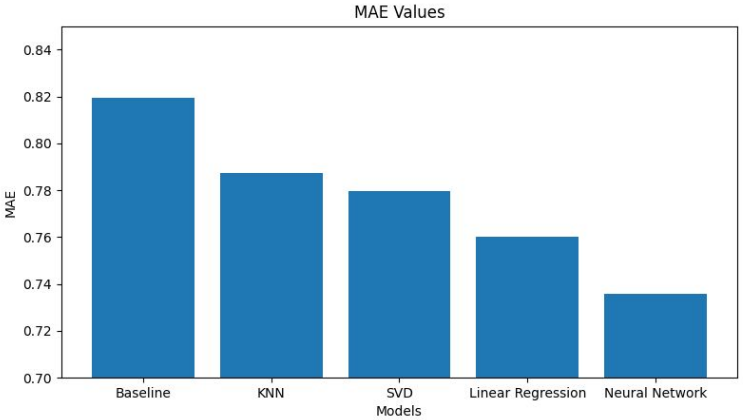
Results Comparison

Model	RMSE	MAE
Baseline	1.0268	0.8193
KNN	0.9959	0.7874
SVD	0.9873	0.7797
Linear Regression	0.9612	0.7602
Neural Network	**0.9384**	**0.7329**

Results Discussion

- All models show improvement from the simple average baseline, showing there are patterns that can be extracted.
- Linear regression worked better than expected for the traditional models, since it utilized supervised learning.
- The neural network performed best, showing that the combination of user demographics along with the genre found more patterns.





Future Improvements

- **Early Stopping**
  - The neural network training shows the potential for overfitting, implementing early stopping would find a spot where the model generalizes best, before it starts overfitting.
- **Dropout**
  - Implementing this into the neural network would allow it to practice without using every known connection, meaning it could be better at predicting unseen data.
- **Rebalancing the Rating Distributions**
  - The MovieLens100K dataset has a larger proportion of high ratings to low ones, and incorporating some synthetic points or rebalancing the data could help the model treat those lower ratings more fairly.

Conclusions

- All advanced machine learning models outperformed the average baseline.
- KNN and SVD, both unsupervised models, were outperformed by linear regression, a supervised learning model.
- The neural network achieved the best performance, likely due to incorporating more user and movie features into the prediction.
- Given more time, further feature engineering and model tuning could improve accuracy.

Acknowledgements

MovieLens 100K Dataset
- Source of the data used.

Pandas documentation
- For data processing.

Scikit-learn documentation
- For model training and metrics.

TensorFlow and Keras documentation
- Used for the neural network model.

CSCI 335/635 Course Slides
- For general reference material.